VIVEK RAJU KOSIGI

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SUMMARY

VLSI Physical Design Engineer with hands-on RTL-to-GDSII implementation experience on RISC-V processor and NPU subsystems using Synopsys ICC2, Design Compiler, and Cadence Genus, Innovus, and Tempus. Experienced in synthesis, floorplanning, power planning, CTS, routing, timing closure, and signoff through the MeitY C2S ASIC program.

TECHNICAL SKILLS

- **EDA Tools:** Cadence Innovus, Genus, Tempus; Synopsys ICC2, Design Compiler
- **Physical Design Flow:** Synthesis, Floorplanning, Power Planning (PDN), Placement, CTS, Routing, STA, Timing Closure, Signoff (DRC/LVS)
- **Scripting:** TCL (flow automation, reporting), Verilog (RTL, testbench)
- **Concepts:** Static Timing Analysis, Clock Tree Synthesis, IR Drop Analysis, Low-Power Design, SDC Constraints
- **Environment:** Linux, Git

EXPERIENCE

Graduate Apprentice | DLRL (DRDO), Hyderabad — Present

- Preparing technical platform reports on FPGA and embedded AI hardware architectures for defence electronics applications.

RTL-to-GDSII VLSI Engineer (Intern) | C2S Programme, MeitY, Government of India — 2025

- Executed the complete RTL-to-GDSII flow for an 8-bit RISC-V processor subsystem, including RTL simulation, synthesis (Genus), formal verification, physical design (Innovus), STA (Tempus), and GDSII signoff at 90 nm.
- Delivered a 202-cell, 1,517.58 um2 post-route design at 79.82% placement density with zero DRC violations and zero connectivity errors across 219 nets.
- Achieved a +2 ps worst-case hold slack on a single-clock-domain design; the compact design required no additional clock-tree buffering.
- Achieved 0.201 mW total post-route power (56% internal, 36% switching, 8% leakage) at 1.1 V nominal supply.

VLSI Physical Design Trainee | VLSIExpert (Industry Training) — Jun 2024 - Dec 2024

- Completed 6-month hands-on RTL-to-GDSII training on Synopsys Design Compiler and ICC2 in a Linux environment, implementing the full backend flow - synthesis, floorplanning, power planning, placement, CTS, and routing - on digital training designs (4-bit full adder, ALU).
- Performed timing analysis and congestion debugging using ICC2 and Design Compiler concepts; developed TCL automation scripts for flow execution and report generation.

PROJECTS

RV32I + Decoupled NPU ASIC - M.E. Thesis Project — 2024-2025

- Designed and implemented RTL-to-GDSII of a RISC-V processor integrated with a decoupled NPU (command-queue dispatch, ping-pong buffering, 4x MAC banks, INT8, zero-skipping) using Cadence Genus and Innovus at 90 nm, with technology scaling analysis across 45/90/180nm.
- Achieved 1.6 GMAC/s throughput at 184 GMAC/s/W efficiency and 8 cycles/inference, with 7,431 cells, 91,730 um2 core area, and 8.69 mW total power (0.695 nJ/inference) at 90 nm.
- Benchmarked design against ~10 prior works.

Digital IP Blocks - RTL-to-GDSII Physical Design — 2024

- Designed RTL and executed the full backend flow (synthesis, placement, CTS, routing, signoff) for foundational digital IP blocks - ALU, counters, encoders, and MUX - optimizing for power, performance, and area.

FPGA Prototyping - RISC-V + Hardware Accelerator (Artix-7) — 2024

- Implemented a RISC-V processor with MAC and Perceptron hardware accelerators on an Artix-7 FPGA, achieving 100 MHz operating frequency at 0.071 W power, with LCD output for functional verification.
- Interfaced an ultrasonic sensor in Verilog for peripheral-level sensor integration on the same platform.

EDUCATION

M.E. in Embedded & VLSI Design, Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad — 2024 - 2026

CGPA: 8.3

B.Tech in Electronics & Communication Engineering, K.L. University, Hyderabad — 2020 - 2024

CGPA: 8.82

ACHIEVEMENTS & CERTIFICATIONS

- Published research paper — *"Design & FPGA Implementation of 32-bit RISC-V Processor with MAC & Perceptron Accelerators,"* ICSSIT 2026, SR University, July 2026.
- Published at ICIMIA 2025 - "Design and Simulation of 4-bit Full Adder using DSCH & Microwind"
- Book Chapter (Ongoing) - VLSI Interconnects - IEEE, 2025
- Government of India C2S Internship - MeitY-funded RTL-to-GDSII programme, competitive selection, 2025
- VLSIExpert ID 536 - 6-month industry Physical Design training (Jun-Dec 2024)